CIRCULAR ECONOMY, KEY SECTORS

India's automotive, construction, agricultural, and electronics sectors are among the important industries that require the quick adoption of circular economy business models. The industries that create the most jobs are construction and agriculture, with growth in electronics and construction predicted to be the greatest. India is a desirable place for circular concepts and enablers like asset tracking and sharing because of its increasing urbanisation and burgeoning smartphone market.



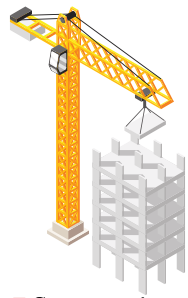
Automotive sector

A McKinsey report suggests India's automobile parts industry could gain a $3.8–5 billion market opportunity in component circularity by FY30. This practice could boost top lines, reduce costs, reduce virgin material use, and decrease carbon emissions.



Electronic and electrical equipment

India faces a growing issue of e-waste due to its expanding economy and large electronic device population. The lack of infrastructure and recycling facilities worsens environmental and health hazards. India's circular electronics policy framework and eco-design standards are being implemented, but a fact-based consultation on economic, social, and environmental impacts is needed.



Construction

India's construction sector contributes to 37% of global energy-related CO2 emissions, with 69% originating from existing buildings. By 2050, outdated methods could account for over 70% of CO2 emissions. Trends in 2024 aim to reshape the industry for an eco-friendly future. Current trends in this direction are green building materials and techniques, the integration of renewable energy, advanced materials, and 3D printing. The construction industry is embracing sustainability, demonstrating innovation and efficiency. India's commitment to sustainable construction has significant environmental impact and resource conservation potential, with the right awareness and education.



Agriculture

The Indian organic food market has a projected 24% CAGR from 2021 to 2026. The pandemic has highlighted the fragility of a linear food system, leading to a 51% YoY increase in exports of organic food products. The Indian F&B industry should leverage these opportunities to build a regenerative food and agricultural system involving farm-level practices, crop rotation, and sustainable packaging. The government has also taken initiatives to support the circular economy.

CIRCULARITY EVALUATION AT THE CORPORATE LEVEL

Biogas Forum India (BiGFIN), a technical society promoting biogas and biofertilizer in India, aims to promote the circular economy and reduce waste generation. It focuses on resource efficiency, waste disposal, and utilisation and has the potential to help build sustainable societies. Biogas-based energy can be generated from organic waste, such as cattle dung, agricultural residues, food waste, and MSW, and can be used in vehicular applications in transportation. BiGFIN works towards the systematic implementation of biogas programmes and contributes to capacity building at all levels. The organisation is actively involved in policy advocacy and capacity building in the biogas domain. The integration of bioenergy programmes like Sustainable Alternative Towards Affordable Transportation (SATAT) and Galvanising Organic Bio-Agro Resources (OBAR) with biogas is a promising scenario for the compressed biogas industry.

PROF. VIRENDRA KUMAR VIJAY,
General Secretary, Biogas Forum India

Johnson Controls (JCI), a global leader in smart, healthy, and sustainable buildings, is transforming indoor spaces with AI and data-driven solutions. Prioritising energy efficiency, waste reduction, and resource conservation, JCI contributes to a circular economy and a net-zero future, employing over 100,000 employees across 150 countries.

CERO (Mahindra MSTC Recycling Pvt Ltd., a joint venture between Mahindra Accelo Ltd. and MSTC Ltd., a GoI enterprise), is determined to realise the true potential of the circular economy. Having set up India's largest network of recycling facilities in India (presence in 41 cities) and recycled more than 21,000 vehicles till date, CERO has contributed immensely to the circular economy by re-inducting steel, plastic, copper, aluminium, rubber, etc. back into the economy. Adoption of a scrappage policy will play an important role in recycling industry formalisation and realising the vision of a circular economy.

MR. SUMIT ISSAR
(MD, Mahindra Accelo, Director, Cero)

Earth Tatva uses TatvaMix, a proprietary recycled ceramic material, to reduce mining for natural resources by up to 60%. Ceramic manufacturing industries face challenges in disposing of landfilled production rejects due to their non-biodegradable nature, as demonstrated by archaeology. TatvaMix is stronger, recyclable, energy-efficient, food-safe, microwave- and dishwasher-friendly, and offers 22.4% lower carbon emissions compared to conventional ceramics.

MR. SHASHANK NIMKAR
(Founder and CEO, Earth Tatva)

Siemens defines its ESG ambitions within the DEGREE sustainability framework—a 360-degree approach reflecting our core sustainability values. By combining the real and digital worlds, we empower our customers and partners to scale their sustainability impact faster across three impact areas: decarbonisation and energy efficiency, resource efficiency and circularity, and people-centricity and societal impact. 90% of Siemens' portfolio enables our customers to have a positive sustainability impact.

Recykal, a Hyderabad-based cleantech startup, is a leader in the circular economy, offering a digitised waste management ecosystem for various categories, channelling 7,00,000 MT of waste and reclaiming 70 thousand metric tonnes of ocean-bound plastic under Project Samudra Manthan.

MR. ABHAY DESHPANDE
founder and CEO